

Data analysis

```
# -----
# Load + prepare data
# -----
truth_ratings <- read_csv("final_successful.csv") %>%
  mutate(
    participant_id      = factor(participant_id),
    repetition_numeric  = as.numeric(repetition_times),
    repetition_factor    = factor(repetition_times, levels = c(0, 1, 9, 18, 27)),
    truth_score         = as.numeric(truth_score)
  )

## Rows: 12800 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): statement_id, participant_id, gender, edu_level
## dbl (5): set_id, cb_version, repetition_times, truth_score, age
## lgl (1): ethnicity
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

rep_levels <- c("0", "1", "9", "18", "27")

# -----
# Helper: within-subject d (paper footnote)
#  $d = |m1 - m2| / \sqrt{s1^2 + s2^2 - 2*r*s1*s2}$ 
# -----
d_within_r <- function(x, y) {
  ok <- complete.cases(x, y)
  x <- x[ok]; y <- y[ok]
  m1 <- mean(x); m2 <- mean(y)
  s1 <- sd(x); s2 <- sd(y)
  r <- cor(x, y)
  denom <- sqrt(s1^2 + s2^2 - 2 * r * s1 * s2)
  abs(m1 - m2) / denom
}

# -----
# 1) Demographics: age summary
# -----
gender_summary <- truth_ratings %>%
  distinct(participant_id, gender) %>%
  count(gender)

gender_summary
```

```
## # A tibble: 4 x 2
##   gender      n
##   <chr>    <int>
## 1 female    94
## 2 male     94
## 3 nonbinary 8
## 4 other     4
```

```
age_summary <- truth_ratings %>%
  distinct(participant_id, age) %>%
  summarise(
    n = n(),
    mean_age = mean(age, na.rm = TRUE),
    sd_age = sd(age, na.rm = TRUE),
    min_age = min(age, na.rm = TRUE),
    max_age = max(age, na.rm = TRUE)
  )
```

```
age_summary
```

```
## # A tibble: 1 x 5
##       n mean_age sd_age min_age max_age
##   <int>   <dbl> <dbl>   <dbl>   <dbl>
## 1   200    26.6  5.04     18     35
```

```
# -----
# 2) Matched-pairs t-test: repeated (collapsed across >0) vs new (0)
# -----
by_participant_cond <- truth_ratings %>%
  group_by(participant_id, repetition_numeric) %>%
  summarise(mean_truth = mean(truth_score, na.rm = TRUE), .groups = "drop")

by_participant_repeated_vs_new <- by_participant_cond %>%
  mutate(item_type = if_else(repetition_numeric == 0, "new", "repeated")) %>%
  group_by(participant_id, item_type) %>%
  summarise(mean_truth = mean(mean_truth, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = item_type, values_from = mean_truth)

tt_repeated_vs_new <- t.test(
  by_participant_repeated_vs_new$repeated,
  by_participant_repeated_vs_new$new,
  paired = TRUE
)

tt_repeated_vs_new
```

```
##
## Paired t-test
##
## data:  by_participant_repeated_vs_new$repeated and by_participant_repeated_vs_new$new
## t = 58.374, df = 199, p-value < 2.2e-16
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
```

```
## 0.9627465 1.0300660
## sample estimates:
## mean difference
## 0.9964062
```

```
d_repeated_vs_new <- d_within_r(
  by_participant_repeated_vs_new$repeated,
  by_participant_repeated_vs_new$new
)

d_repeated_vs_new
```

```
## [1] 4.127698
```

```
msd_repeated_new <- by_participant_repeated_vs_new %>%
  summarise(
    M_repeated = mean(repeated, na.rm = TRUE),
    SD_repeated = sd(repeated, na.rm = TRUE),
    M_new = mean(new, na.rm = TRUE),
    SD_new = sd(new, na.rm = TRUE)
  )

msd_repeated_new
```

```
## # A tibble: 1 x 4
##   M_repeated SD_repeated M_new SD_new
##   <dbl>      <dbl> <dbl> <dbl>
## 1     4.64      0.161  3.64  0.168
```

```
# -----
# 3) Per-participant correlations: linear vs log (reps + 1 for both)
# -----

wide_means <- by_participant_cond %>%
  mutate(rep_chr = as.character(repetition_numeric)) %>%
  select(participant_id, rep_chr, mean_truth) %>%
  pivot_wider(names_from = rep_chr, values_from = mean_truth)

stopifnot(all(rep_levels %in% names(wide_means)))

rep_num <- as.numeric(rep_levels) + 1
rep_log <- log(rep_num)

corrs <- wide_means %>%
  rowwise() %>%
  mutate(
    r_linear = {
      y <- c_across(all_of(rep_levels))
      if (anyNA(y)) NA_real_ else cor(y, rep_num)
    },
    r_log = {
      y <- c_across(all_of(rep_levels))
      if (anyNA(y)) NA_real_ else cor(y, rep_log)
    }
  )
```

```

    }
  ) %>%
  ungroup()

corr_summary <- corrs %>%
  summarise(
    mean_r_linear = mean(r_linear, na.rm = TRUE),
    sd_r_linear   = sd(r_linear, na.rm = TRUE),
    min_r_linear  = min(r_linear, na.rm = TRUE),
    max_r_linear  = max(r_linear, na.rm = TRUE),
    prop_pos_linear = mean(r_linear > 0, na.rm = TRUE),

    mean_r_log = mean(r_log, na.rm = TRUE),
    sd_r_log   = sd(r_log, na.rm = TRUE),
    min_r_log  = min(r_log, na.rm = TRUE),
    max_r_log  = max(r_log, na.rm = TRUE),
    prop_pos_log = mean(r_log > 0, na.rm = TRUE)
  )

as.data.frame(corr_summary)

##   mean_r_linear sd_r_linear min_r_linear max_r_linear prop_pos_linear
## 1      0.6768811  0.2003865  0.0556757   0.9756884             1
##   mean_r_log   sd_r_log min_r_log max_r_log prop_pos_log
## 1  0.8149456 0.1512535 0.1641655 0.9987703             1

# Number of participants with r_linear > 0
n_linear_pos <- sum(corrs$r_linear > 0, na.rm = TRUE)

# Total valid participants (non-missing r_linear)
n_total <- sum(!is.na(corrs$r_linear))

# Proportion (often reported in papers)
prop_linear_pos <- n_linear_pos / n_total

n_linear_pos

## [1] 200

n_total

## [1] 200

prop_linear_pos

## [1] 1

tt_corr <- t.test(abs(corrs$r_log), abs(corrs$r_linear), paired = TRUE)
tt_corr

```

```
##
## Paired t-test
##
## data: abs(corrs$r_log) and abs(corrs$r_linear)
## t = 22.042, df = 199, p-value < 2.2e-16
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.1257127 0.1504164
## sample estimates:
## mean difference
## 0.1380645
```

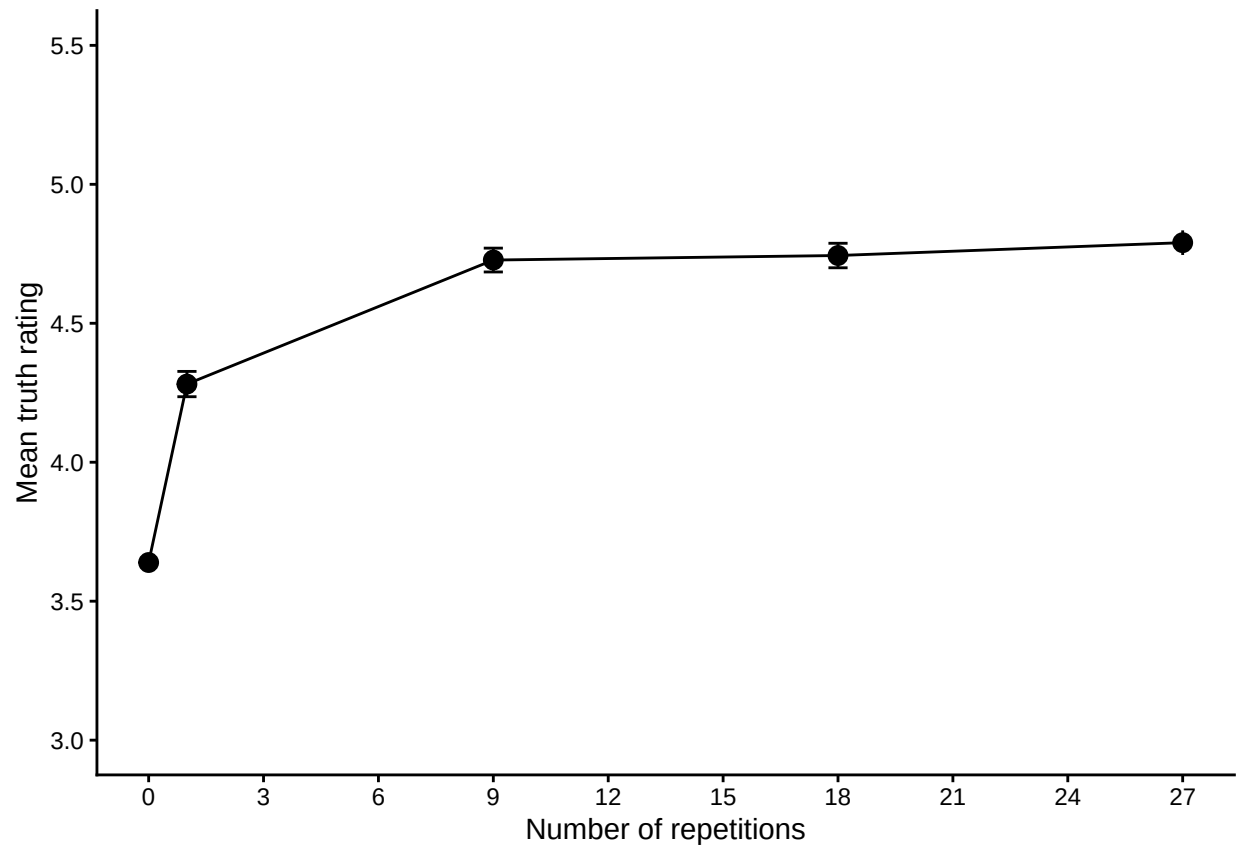
```
diff_r <- abs(corrs$r_log) - abs(corrs$r_linear)
d_dz_corr <- mean(diff_r, na.rm = TRUE) / sd(diff_r, na.rm = TRUE)
d_dz_corr
```

```
## [1] 1.558598
```

```
##
# graph
##

# Compute means + CI using numeric repetitions
means_plot <- truth_ratings %>%
  group_by(repetition_numeric) %>%
  summarise(
    mean_truth = mean(truth_score, na.rm = TRUE),
    se = sd(truth_score, na.rm = TRUE) / sqrt(n()),
    ci95 = qt(0.975, df = n() - 1) * se,
    .groups = "drop"
  )

# Plot with numeric scale + regular ticks
ggplot(means_plot, aes(x = repetition_numeric, y = mean_truth)) +
  geom_point(size = 3) +
  geom_line() +
  geom_errorbar(aes(ymin = mean_truth - ci95,
                    ymax = mean_truth + ci95,
                    width = 0.5)) +
  scale_x_continuous(
    breaks = seq(0, 27, by = 3),
    limits = c(0, 27)
  ) +
  coord_cartesian(ylim = c(3, 5.5)) + # update this when changing variation
  labs(
    x = "Number of repetitions",
    y = "Mean truth rating"
  ) +
  theme_classic()
```



```
# -----
# 4) RM ANOVA + Bonferroni pairwise comparisons
# -----
aov_rm <- aov_car(
  truth_score ~ repetition_factor + Error(participant_id / repetition_factor),
  data = truth_ratings,
  factorize = FALSE
)
```

```
## Warning: More than one observation per design cell, aggregating data using 'fun_aggregate = mean'.
## To turn off this warning, pass 'fun_aggregate = mean' explicitly.
```

```
aov_rm
```

```
## Anova Table (Type 3 tests)
##
## Response: truth_score
##           Effect             df  MSE          F ges p.value
## 1 repetition_factor 3.61, 717.52 0.09 568.15 *** .696  <.001
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
##
## Sphericity correction method: GG
```

```

emm_rep <- emmeans(aov_rm, ~ repetition_factor)

pairs_bonf <- pairs(emm_rep, adjust = "bonferroni") %>%
  summary(infer = TRUE) %>%
  as.data.frame()

pairs_bonf

## contrast      estimate      SE df lower.CL upper.CL t.ratio p.value
## X0 - X1      -0.6420313 0.02608625 199 -0.7160816 -0.5679809 -24.612 <.0001
## X0 - X9      -1.0882813 0.02546910 199 -1.1605797 -1.0159828 -42.729 <.0001
## X0 - X18     -1.1045313 0.02665826 199 -1.1802053 -1.0288572 -41.433 <.0001
## X0 - X27     -1.1507813 0.02426733 199 -1.2196683 -1.0818942 -47.421 <.0001
## X1 - X9      -0.4462500 0.02934200 199 -0.5295424 -0.3629576 -15.209 <.0001
## X1 - X18     -0.4625000 0.03433000 199 -0.5599517 -0.3650483 -13.472 <.0001
## X1 - X27     -0.5087500 0.02970106 199 -0.5930616 -0.4244384 -17.129 <.0001
## X9 - X18     -0.0162500 0.03241618 199 -0.1082690 0.0757690 -0.501 1.0000
## X9 - X27     -0.0625000 0.03141602 199 -0.1516798 0.0266798 -1.989 0.4802
## X18 - X27    -0.0462500 0.02989605 199 -0.1311151 0.0386151 -1.547 1.0000
##
## Confidence level used: 0.95
## Conf-level adjustment: bonferroni method for 10 estimates
## P value adjustment: bonferroni method for 10 tests

pairs_table <- as.data.frame(pairs_bonf) %>%
  filter(contrast %in% c(
    "X0 - X1",
    "X1 - X9",
    "X9 - X18",
    "X18 - X27"
  )) %>%
  mutate(
    contrast = str_replace(contrast, " - ", " vs "),
    contrast = str_replace_all(contrast, "X", ""),
    estimate = round(abs(estimate), 2),

    # significance flag
    sig = p.value < 0.05,

    # only format the estimate column
    estimate = ifelse(sig,
                      paste0("***", estimate, "***", "***"),
                      as.character(estimate))
  ) %>%
  select(`Repetition Comparisons` = contrast,
         `Mean Difference` = estimate)

kable(
  pairs_table,
  caption = "Mean differences in truth ratings across repetition conditions",
  escape = FALSE
)

```

Table 1: Mean differences in truth ratings across repetition conditions

Repetition Comparisons	Mean Difference
0 vs 1	0.64*
1 vs 9	0.45*
9 vs 18	0.02
18 vs 27	0.05

```
# Build robust numeric p-value lookup for (a,b) keys like "0||1"
pairs_bonf2 <- pairs_bonf %>%
  tidyr::extract(
    contrast,
    into = c("a_raw", "b_raw"),
    regex = "^\\s*(.*?)\\s*-\\s*(.*?)\\s*$",
    remove = FALSE
  ) %>%
  mutate(
    a = stringr::str_extract(a_raw, "\\d+"),
    b = stringr::str_extract(b_raw, "\\d+"),
    key = paste(a, b, sep = "||"),
    p_num = as.numeric(p.value)
  )

p_lookup <- setNames(pairs_bonf2$p_num, pairs_bonf2$key)

get_p_bonf <- function(a, b, lookup) {
  key1 <- paste(a, b, sep = "||")
  key2 <- paste(b, a, sep = "||")

  p1 <- lookup[key1]
  if (length(p1) == 1 && !is.na(p1)) return(unname(p1))

  p2 <- lookup[key2]
  if (length(p2) == 1 && !is.na(p2)) return(unname(p2))

  NA_real_
}

# -----
# 5) Table 2-style output (M/SD header + triangular d matrix)
# -----
desc_by_cond <- truth_ratings %>%
  group_by(repetition_factor) %>%
  summarise(
    M = mean(truth_score, na.rm = TRUE),
    SD = sd(truth_score, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(repetition_factor = as.character(repetition_factor))

msd_header <- desc_by_cond %>%
  mutate(msd = sprintf("\\makecell{M = %.2f \\\\ SD = %.2f}", M, SD)) %>%
```



```

select(repetition_factor, msd) %>%
pivot_wider(names_from = repetition_factor, values_from = msd) %>%
as.data.frame()

cell_matrix <- matrix("",
  nrow = length(rep_levels),
  ncol = length(rep_levels),
  dimnames = list(rep_levels, rep_levels)
)

for (i in seq_along(rep_levels)) {
  for (j in seq_along(rep_levels)) {
    if (i == j) {
      cell_matrix[i, j] <- "-"
    } else if (i < j) {
      a <- rep_levels[i]
      b <- rep_levels[j]

      d_val <- d_within_r(wide_means[[a]], wide_means[[b]])
      p_val <- get_p_bonf(a, b, p_lookup)
      is_sig <- !is.na(p_val) && p_val < 0.05

      cell_txt <- sprintf("d = %.2f%s", d_val, if (is_sig) "*" else "")
      if (is_sig) cell_txt <- paste0("\\textbf{", cell_txt, "}")

      cell_matrix[i, j] <- cell_txt
    }
  }
}

d_table_formatted <- as.data.frame(cell_matrix)

final_table <- bind_rows(
  tibble(` ` = "Ms and SDs of truth ratings") %>% bind_cols(msd_header),
  tibble(` ` = rownames(d_table_formatted)) %>% bind_cols(d_table_formatted)
) %>%
  mutate(across(everything(), as.character))

knitr::kable(final_table, format = "latex", booktabs = TRUE,
  escape = FALSE, align = "lcccc") %>%
  kableExtra::kable_styling(latex_options = c("hold_position")) %>%
  kableExtra::add_header_above(c(" " = 1, "Number of Session 1 repetitions" = 5))

```

	Number of Session 1 repetitions				
	0	1	9	18	27
Ms and SDs of truth ratings	M = 3.64 SD = 0.93	M = 4.28 SD = 0.93	M = 4.73 SD = 0.87	M = 4.74 SD = 0.90	M = 4.79 SD = 0.90
0	–	d = 1.74*	d = 3.02*	d = 2.93*	d = 3.35*
1		–	d = 1.08*	d = 0.95*	d = 1.21*
9			–	d = 0.04	d = 0.14
18				–	d = 0.11
27					–